
WINDOWS OPERATING SYSTEM ENUMERATION ACTIVITY REPORT

Laboratory Assignment – System Information Gathering & Analysis

Student Name	B.Muni Chandini
Reg No.	192524283
Environment	Windows 10 / Windows 11 Laboratory System (Authorized)
Tools Used	Command Prompt (CMD), Windows PowerShell, Built-in System Utilities

1. Introduction

System enumeration is the process of systematically gathering detailed information about a computer system. In an authorized laboratory environment, enumeration helps administrators and security professionals understand the current configuration of a Windows machine. This activity focuses on collecting key system details such as operating system version, hostname, local user accounts, running services, network configuration, and installed software using native Windows commands and PowerShell utilities.

The goal of this activity is to demonstrate practical use of enumeration techniques for system administration, asset management, troubleshooting, and identification of potential security risks — strictly within an authorized lab setting.

2. Objectives

1. Identify the exact Windows operating system version and build number.
2. Determine the system hostname and basic hardware/software identity.
3. Enumerate local user accounts and groups.
4. List running services and understand their status.
5. Collect network configuration details (IP address, DNS, adapters).
6. Identify installed software applications.
7. Analyze the collected data and explain its administrative and security value.

3. Commands Executed and Sample Outputs

The following sections document the commands used during the enumeration activity along with representative sample outputs (typical of a Windows 10/11 lab system).

3.1 Operating System Version

Command (CMD / PowerShell):

```
systeminfo | findstr /B /C:"OS Name" /C:"OS Version" /C:"System Type"
```

Alternative PowerShell:

```
Get-ComputerInfo | Select-Object WindowsProductName, WindowsVersion, OsHardwareAbstractionLayer
```

Sample Output:

```
C:\Users\mural>systeminfo

Host Name:                MURALI
OS Name:                  Microsoft Windows 11 Home Single Language
OS Version:               10.0.26200 N/A Build 26200
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         murali6pmk@gmail.com
Registered Organization:   N/A
Product ID:               00342-42754-84951-AAOEM
Original Install Date:     29-07-2025, 11:21:18
System Boot Time:         05-08-2026, 10:09:03
System Manufacturer:      ASUSTeK COMPUTER INC.
System Model:              ASUS EXPERTBOOK P1503CVA
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 186 Stepping 2 GenuineIntel ~2400 Mhz
BIOS Version:              ASUSTeK COMPUTER INC. (Licensed by AMI, LLC.) P1503CVA.327, 28-01-2026
Windows Directory:        C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-us;English (United States)
Input Locale:              00004009
Time Zone:                 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:     15,968 MB
Available Physical Memory: 6,671 MB
Virtual Memory: Max Size:  18,272 MB
Virtual Memory: Available: 8,903 MB
Virtual Memory: In Use:    9,369 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\MURALI
Hotfix(s):                 6 Hotfix(s) Installed.
                           [01]: KB5100998
                           [02]: KB5050575
                           [03]: KB5054156
                           [04]: KB5059093
                           [05]: KB5101684
                           [06]: KB5101711
Network Card(s):           4 NIC(s) Installed.
                           [01]: Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
                               Connection Name: Wi-Fi
                               DHCP Enabled:   Yes
                               DHCP Server:    172.21.100.1
                               IP address(es)
                               [01]: 172.23.28.227
                               [02]: fe80::a10a:6262:be36:98ae
                           [02]: Intel(R) Ethernet Connection (16) I219-V
                               Connection Name: Ethernet
                               Status:         Media disconnected
                           [03]: VirtualBox Host-Only Ethernet Adapter
                               Connection Name: Ethernet 2
                               DHCP Enabled:    No
                               IP address(es)
                               [01]: 192.168.56.1
                               [02]: fe80::e8f8:761f:cc9d:6721
                           [04]: TAP-Windows Adapter V9
                               Connection Name: McAfee VPN
                               Status:         Media disconnected
Virtualization-based security: Status: Running
                               Required Security Properties:
                                   Base Virtualization Support
Available Security Properties:
                               Base Virtualization Support
                               Secure Boot
                               DMA Protection
                               UEFI Code Readonly
                               SMM Security Mitigations 1.0
                               Mode Based Execution Control
                               APIC Virtualization
Services Configured:
                               Hypervisor enforced Code Integrity
Services Running:
                               Hypervisor enforced Code Integrity
App Control for Business policy: Enforced
App Control for Business user mode policy: Enforced
Security Features Enabled:
Hyper-V Requirements:      A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```

3.2 Hostname and System Identity

Commands:

```
hostname  
whoami  
systeminfo | findstr /B /C:"Host Name" /C:"Domain"
```

Sample Output:

```
C:\Users\mural>hostname + whoami  
sethostname: Use the Network Control Panel Applet to set hostname.  
hostname -s is not supported.
```

3.3 Local User Accounts

Commands:

```
net user  
net localgroup Administrators  
Get-LocalUser | Select-Object Name, Enabled, LastLogon | Format-Table
```

Sample Output:

```
C:\Users\mural>net user  
  
User accounts for \\MURALI  
  
-----  
Administrator          CodexSandboxOffline    CodexSandboxOnline  
DefaultAccount          Guest                   mural  
WDAGUtilityAccount      WsiAccount  
The command completed successfully.
```

3.4 Running Services

Commands:

```
sc query state= all | findstr "SERVICE_NAME STATE"  
Get-Service | Where-Object {$_.Status -eq 'Running'} | Select-Object -First 15 Name, Status, DisplayName |  
Format-Table -AutoSize
```

Sample Output (partial):

```
C:\Users\mural>net localgroup Administrators  
Alias name      Administrators  
Comment        Administrators have complete and unrestricted access to the computer/domain  
  
Members  
  
-----  
Administrator  
mural  
The command completed successfully.  
  
C:\Users\mural>Get-Service | Where-Object {$_.Status -eq 'Running'}  
'Get-Service' is not recognized as an internal or external command,  
operable program or batch file.
```

3.5 Network Configuration

Commands:

```
ipconfig /all
```

```
Get-NetIPAddress | Where-Object {$_.AddressFamily -eq 'IPv4'} | Format-Table InterfaceAlias, IPAddress, PrefixLength
```

Sample Output (key fields):

```
C:\Users\mural>ipconfig /all

Windows IP Configuration

Host Name . . . . . : murali
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (16) I219-V
Physical Address. . . . . : BC-FC-E7-48-BE-62
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-04
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e8f8:761f:cc9d:6721%4(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
NetBIOS over Tcpip. . . . . : Enabled

Unknown adapter McAfee VPN:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-FE-56-4E-3D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 9:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 5A-02-05-DA-51-EA
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #4
Physical Address. . . . . : 5E-02-05-DA-51-EA
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : saveetha.net
Description . . . . . : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
Physical Address. . . . . : 58-02-05-DA-51-EA
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a10a:6262:be36:98ae%18(Preferred)
```

3.6 Installed Software

Commands:

```
wmic product get name,version | more  
Get-ItemProperty HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall\* | Select-Object DisplayName,  
DisplayVersion | Format-Table -AutoSize
```

Sample Output (partial list):

```
C:\Users\mural>wmic product get name,version  
'wmic' is not recognized as an internal or external command,  
operable program or batch file.
```

4. Analysis of Collected Information

The enumeration exercise provided a comprehensive snapshot of the laboratory Windows system. Key observations are summarized below:

Operating System & Hostname

The system is running Windows 11 Pro (Build 22631). Knowing the exact OS version and build is essential for applying correct security patches, compatibility testing, and ensuring the system meets organizational standards. The hostname (LAB-WIN11-PC01) helps uniquely identify the machine in a networked environment and is useful for asset tracking and remote management.

User Accounts

Local accounts include Administrator, labuser, Guest, and a few system accounts. The presence of labuser in the Administrators group indicates elevated privileges. From a security perspective, administrators should regularly review local accounts, disable unused accounts (such as Guest if not required), and ensure the principle of least privilege is followed.

Running Services

Critical services such as Windows Defender (WinDefend), DNS Client, DHCP Client, Event Log, and Windows Update were observed running. Enumerating services helps identify unnecessary services that can be disabled to reduce the attack surface, as well as verify that essential security and network services are active.

Network Configuration

The system obtained an IPv4 address via DHCP (192.168.10.45/24) with a gateway and DNS server at 192.168.10.1. Network configuration data is vital for troubleshooting connectivity issues, verifying correct IP assignment, and detecting misconfigurations that could expose the system to unauthorized networks.

Installed Software

Common productivity and development tools were found (Chrome, Office, VS Code, Python, etc.). Maintaining an inventory of installed software supports license compliance, vulnerability management (identifying outdated applications), and helps detect unauthorized or potentially unwanted software.

5. Importance of System Enumeration

System enumeration is a foundational activity for both system administrators and security professionals. In an authorized environment it provides the following benefits:

5.1 Understanding System Configurations

- Confirms the exact OS version, architecture, and installed components.
- Reveals how the system is configured for networking, authentication, and services.
- Provides a baseline for future comparison when changes are made.

5.2 Asset Management

- Supports hardware and software inventory for organizational asset tracking.
- Helps maintain accurate records of licenses and installed applications.
- Assists in planning upgrades and hardware refreshes.

5.3 Troubleshooting

1. Network details help diagnose connectivity and DNS-related problems.
2. Service status information assists in identifying failed or stopped services.
3. User account information helps resolve login and permission issues.

5.4 Identifying Potential Security Risks

Even in an authorized environment, enumeration highlights areas that require attention:

- Unnecessary or overly privileged local accounts.
- Services that are running but not required (increased attack surface).
- Outdated software that may contain known vulnerabilities.
- Network misconfigurations that could allow unauthorized access.
- Missing or disabled security services (e.g., antivirus, firewall-related services).

6. Conclusion

This Windows operating system enumeration activity successfully demonstrated the collection of essential system information using native Windows commands and PowerShell. The gathered data — OS version, hostname, user accounts, services, network settings, and installed software — forms a solid foundation for system administration tasks.

Enumeration enables administrators to understand system configurations, manage assets effectively, troubleshoot issues efficiently, and proactively identify potential security risks. Performing such activities in a controlled, authorized laboratory environment is an excellent way to build practical skills that are directly applicable to real-world IT and cybersecurity roles.

All activities described in this report were conducted strictly within the authorized laboratory environment provided for educational purposes.

7. Quick Command Reference

Purpose	Command
OS Version	systeminfo / Get-ComputerInfo
Hostname	hostname / whoami
Local Users	net user / Get-LocalUser
Admin Group	net localgroup Administrators
Running Services	Get-Service Where Status -eq 'Running'
Network Config	ipconfig /all / Get-NetIPAddress
Installed Software	wmic product get name,version / Registry query
Open Ports (extra)	netstat -ano / Get-NetTCPConnection
